Department of Computer Science & Engineering
Jahangirnagar University, Savar, Dhaka

PMSCS

Course: Software Project Management

Assignment: Step Wise Project Planning — Applied Case Study

Total Marks	100 (Part A: 40 | Part B: 30 | Part C: 30)
Submission Deadline	19th June as per stated in Classroom
Submission Mode	Soft copy (PDF) submitted to the cclassroom
Group Policy	Individual submission; collaboration is not permitted
Reference	Chapter 2 — Step Wise: An Approach to Planning Software Projects

General Instructions

- Read Chapter 2 of Bob Hughes & Mike Cotterell, Software Project Management (4th Ed.) thoroughly before attempting the assignment.
- Justify all answers with reasoning. Vague or one-line answers will not receive full marks.
- Where diagrams are requested, draw neatly and label all nodes/edges clearly.
- All written answers must be in your own words. Plagiarism will result in zero marks.
- State any assumptions you make explicitly at the beginning of the respective question.

Part A — Conceptual Understanding (50 Marks)

Answer all four questions in this part.

Question 1 [10 Marks]

The Step Wise framework is designed around three key aspirations: practicality, scalability, and range of application.

a) In your own words, explain what is meant by each of the three aspirations and why they matter in real-world software project management. [6 marks]

b) Compare Step Wise with any project management method. In what ways does Step Wise borrow from PRINCE2, and where does it differ? [4 marks]

Question 2 [10 Marks]

Step 1 of Step Wise focuses on establishing project scope and objectives, including stakeholder analysis.

c) Why is it important to distinguish between a project authority and other stakeholders? Use the psychometric-testing scenario from the lecture as an illustrative example. [5 marks]

d) Explain how stakeholder analysis can lead to a revision of project objectives. Give an example of such a revision from the scenario discussed in class. [5 marks]

Question 3 [15 Marks]

Steps 3 through 5 of Step Wise address project characterisation, product/activity identification, and effort estimation.

e) Differentiate between objective-based and product-based projects. Why does this distinction affect the choice of life cycle model? [7 marks]

f) What is a Product Breakdown Structure (PBS)? Construct a simple PBS for an online student-registration system (at least six products). [8 marks]

Question 4 [15 Marks]

Risk management is central to Steps 6 and 7 of Step Wise.

g) Describe the difference between risk reduction and contingency planning. Why must both be addressed in a project plan? [6 marks]

h) For the following risks identified in the class scenario, identify: (i) a risk-reduction activity, and (ii) a contingency action for each. [9 marks]

- Team members refuse to complete psychometric questionnaires.
- Project managers are unwilling to use the application.
- The developer is unfamiliar with the selected implementation technology.

Part B — Applied Problem Solving (50 Marks)

Answer both questions in this part.

Question 5 [25 Marks]

CSE JU decided to develop online based students information system. All activities for studentdent registration, course registration, project distribution, class evaluation, and similar works are to be automated in online web-based system. The platform must integrate with the national mobile payment gateway (e.g., bKash/Nagad) for payment and settlement to Agrani Bank acount. The expected delivery timeline is 9 months with a 12-member cross-functional team.

Apply the Step Wise framework (Steps 0–7) to this scenario. For each step, provide:

- A concise description of what needs to be decided or produced at that step.
- The specific outputs/artefacts you would generate for this project.
- Any challenges particular to this context (fintech, Bangladesh regulatory environment, mobile platform constraints).

Your answer must cover all eight steps (0 through 7) and demonstrate understanding of how each step feeds into the next. A tabular or structured format is acceptable.

Question 6 [25 Marks]

Using the activity network concept from Step 4.4 of Step Wise, answer the following for the digital loan platform project described in Question 5:

i) Identify at least eight distinct activities required to deliver the platform, expressed in the verb + noun form (e.g., "Design result processing model"). Briefly justify each activity. [8 marks]

j) Draw a clearly labelled Activity Network (precedence diagram) showing the dependencies among your identified activities. Indicate the critical path. [10 marks]

k) Construct a Gantt chart (using Gantt-Project) covering the first 12 weeks of the project. Assign at least two named resources and indicate their allocation across activities. [7 marks]

Part C — Critical Analysis & Research (30 Marks) [Bonus Mark]

Answer both questions in this part.

Question 7 [15 Marks]

Step 3 of Step Wise requires selecting a general life cycle approach — waterfall, incremental delivery, evolutionary prototyping, or another model.

l) Critically compare the evolutionary prototype approach with the waterfall model for projects where user requirements are uncertain. Under what conditions would you recommend each, and why? [7 marks]

m) In the class scenario (psychometric testing system), the evolutionary prototype was suggested as a solution. Identify two specific risks in that scenario that make prototyping preferable, and explain how prototyping mitigates those risks. [5 marks]

n) How does the choice of life cycle model interact with the resource-allocation decisions made in Step 7? Discuss with an example. [3 marks]

Question 8 [15 Marks]

Reflect on the Step Wise framework as a whole.

o) Step Wise claims to be scalable — equally useful for small and large projects. Critically evaluate this claim. What adaptations, if any, would you make when applying Step Wise to a large-scale government e-services project in Bangladesh? [8 marks]

p) Modern software development has widely adopted Agile and Scrum. Discuss whether Step Wise is compatible with Agile values. Which steps of Step Wise would you retain, modify, or discard when working in an Agile environment? Justify your answer. [7 marks]

Marking Scheme Summary

Criterion	Marks	Notes
Conceptual accuracy	**30**	Correct understanding of Step Wise concepts,

		terminology, and sequence
Application to scenario	**30**	Demonstrates ability to apply the framework to a realistic project context
Critical thinking & analysis	**25**	Questions 7 and 8 are graded heavily on depth of argument and evidence
Diagrams (PBS, Activity Network, Gantt)	**10**	Clarity, labelling, and correctness of all drawn artefacts
Presentation & referencing	**5**	Logical structure, academic writing quality, and citations where applicable

Note: Marks for each sub-question are indicative. Examiners may adjust within ±2 marks per sub-question at their discretion.

— End of Assignment —